



HK IMO Training 2023-2024 Problem Set 8 Suggested Solutions



Problem 29 (Grand Duchy of Lithuania 2018 Problem 4). Find all positive integers n such that there exists a positive integer k satisfying the following property: for every positive divisor d of n , the integer $d - k$ is also a (not necessarily positive) divisor of n .

Solution. Answer: 1, 4, 6 and any prime number

When $n = 1$, we can take $k = 2$. When $n = 4$, we can take $k = 3$. When $n = 6$, we can take $k = 4$. When n is a prime, we can take $k = n + 1$, since the only positive divisors of n are 1 and n , and we have $1 - k = -n \mid n$ and $n - k = -1 \mid n$.

Now, assume $n \geq 8$ and n is not a prime. Suppose such a k exists.

- If $k < \frac{n}{2}$, consider $n - k \mid n$ (as $n \mid n$). Note that $\frac{n}{2} < n - k < n$. This is impossible as there cannot be any divisor of n between $\frac{n}{2}$ and n .
- If $k \geq \frac{n}{2}$, consider $1 - k \mid n$ (as $1 \mid n$). This implies $k - 1 \mid n$. Note that

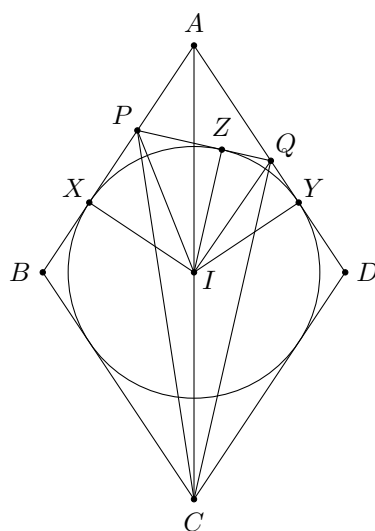
$$k - 1 \geq \frac{n}{2} - 1 > \frac{n}{3}$$

since $n \geq 8$. Therefore, we must have $k - 1 = \frac{n}{2}$ or $k - 1 = n$.

- If $k - 1 = \frac{n}{2}$, consider $n - k \mid n$. This implies $\frac{n}{2} - 1 \mid n = 2\left(\frac{n}{2} - 1\right) + 2$, and hence $\frac{n}{2} - 1 \mid 2$. This is impossible as $\frac{n}{2} - 1 \geq 3$.
- If $k - 1 = n$, consider the largest proper divisor d of n . Note that $d - k \mid n$, which means $d - n - 1 \mid n$, or $n - d + 1 \mid n$.

It follows that $n - d + 1 \mid d - 1$, and so $n - d + 1 \leq d - 1$ (as $d \neq 1$). This yields $d \geq \frac{n+2}{2}$. But then $\frac{n}{2} < d < n$, which is impossible. □

Problem 30 (Canada MO 2020 Problem 2). Let $ABCD$ be a rhombus. Consider two movable points P and Q on the sides AB and AD respectively, such that PQ is tangent to the inscribed circle of $ABCD$. Prove that the area of $\triangle CPQ$ is a constant independent of the positions of P and Q .



Solution. Let I be the centre of the incircle. Note that I is the midpoint of AC and BD . Let X, Y, Z be the contact points of the incircle and AB, AD, PQ respectively. Using $IA = IC$, $PZ = PX$ and $QZ = QY$, we have

$$[CPQ] = [ICP] + [ICQ] + [IPZ] + [IQZ] = [IAP] + [IAQ] + [IPX] + [IQY] = [AXIY].$$

This is a fixed constant independent of P and Q . □

Problem 31 (Canada MO 2021 Problem 3). Let $n \geq 4$ be a positive integer. There are n boys and n girls sitting around a circular table. Two girls will chat with one another if one of the following holds:

- there is at most one person sitting between them; or
- there are exactly two people sitting between them, at least one of whom is a boy.

Prove that there are at least n pairs of girls who will chat with one another no matter how the people are seated.

Solution. Let $G_1, G_2, \dots, G_n$ be the girls in clockwise order, and let $a_i \geq 0$ be the number of boys between G_i and G_{i+1} (where the index is taken modulo n). Note that G_i chats with G_{i+1} iff $a_i \leq 2$, while G_i chats with G_{i+2} iff $a_i + a_{i+1} \leq 1$.

Let S be the set of all indices i such that $a_i \geq 3$, and let $k = |S|$. Since $a_1 + a_2 + \dots + a_n = n$, we must have $k \leq \frac{n}{3}$. There are $n - k$ pairs of G_i and G_{i+1} who will chat. Also, consider

$$\sum_{\substack{i=1 \\ i, i+1 \notin S}}^n (a_i + a_{i+1}) \leq 2 \sum_{i=1}^n a_i - 2 \sum_{\substack{i=1 \\ i \in S}}^n a_i \leq 2n - 6k.$$

The leftmost sum consists of $m \geq n - 2k$ terms of the form $a_i + a_{i+1}$. The above inequality shows that there are at most $\frac{2n - 6k}{2} = n - 3k$ terms which are at least 2. Therefore, at least $m - (n - 3k) \geq k$ terms are at most 1. Correspondingly, at least k pairs G_i and G_{i+2} will chat. It follows that there are at least $(n - k) + k = n$ pairs of girls who will chat. $\square$

Problem 32 (Romania TST 2014 Day 3 Problem 3/Hardy's inequality). Find the smallest real number c such that

$$\sum_{k=1}^n \left(\frac{1}{k} \sum_{j=1}^k x_j \right)^2 \leq c \sum_{k=1}^n x_k^2$$

for all positive integers n and positive real numbers $x_1, x_2, \dots, x_n$.

Solution. Answer: 4.

Let $s_k = x_1 + x_2 + \dots + x_k$ and $t_k = \frac{s_k}{k}$ for $k = 0, 1, \dots, n$ (where $s_0 = 0$). We first prove that

$$\sum_{k=1}^n \frac{s_k^2}{k^2} \leq 2 \sum_{k=1}^n \frac{x_k s_k}{k}. \quad (1)$$

Indeed, note that

$$\begin{aligned} \frac{s_k^2}{k^2} - \frac{2x_k s_k}{k} &= t_k^2 - 2(s_k - s_{k-1})t_k \\ &= t_k^2 - 2(kt_k - (k-1)t_{k-1})t_k \\ &= (1-2k)t_k^2 + 2(k-1)t_{k-1}t_k \\ &\leq (1-2k)t_k^2 + (k-1)(t_{k-1}^2 + t_k^2) \\ &= (k-1)t_{k-1}^2 - kt_k^2. \end{aligned}$$

Therefore,

$$\sum_{k=1}^n \left(\frac{s_k^2}{k^2} - \frac{2x_k s_k}{k} \right) \leq \sum_{k=1}^n [(k-1)t_{k-1}^2 - kt_k^2] = -nt_n^2 \leq 0.$$

This proves (1). By the Cauchy-Schwarz inequality, we have

$$\left(\sum_{k=1}^n \frac{s_k^2}{k^2} \right)^2 \leq 4 \left(\sum_{k=1}^n \frac{x_k s_k}{k} \right)^2 \leq 4 \left(\sum_{k=1}^n x_k^2 \right) \left(\sum_{k=1}^n \frac{s_k^2}{k^2} \right).$$

This implies

$$\sum_{k=1}^n \frac{s_k^2}{k^2} \leq 4 \sum_{k=1}^n x_k^2.$$

Therefore, the inequality holds when $c = 4$.

On the other hand, consider the sequence $x_k = \frac{1}{\sqrt{k}}$. Then we have

$$\sum_{j=1}^k x_j = \sum_{j=1}^k \frac{1}{\sqrt{j}} \geq \int_1^{k+1} \frac{1}{\sqrt{t}} dt = 2(\sqrt{k+1} - 1),$$

and hence

$$\sum_{k=1}^n \left(\frac{1}{k} \sum_{j=1}^k x_j \right)^2 \geq 4 \sum_{k=1}^n \frac{k+2-2\sqrt{k+1}}{k^2} > 4 \left(\sum_{k=1}^n \frac{1}{k} - c_1 \right)$$

for some constant c_1 (as both infinite sums of k^{-2} and $k^{-\frac{3}{2}}$ converge). Since the harmonic sum $\sum_{k=1}^{\infty} \frac{1}{k}$ diverges, for each $\varepsilon > 0$, this is larger than

$$(4 - \varepsilon) \sum_{k=1}^n \frac{1}{k} = (4 - \varepsilon) \sum_{k=1}^n x_k^2$$

for all sufficiently large n . Therefore, c cannot be smaller than 4. This shows 4 is the best constant. □